ENGLISH PROJECT REPORT VERSION

Drone Project - Third Progress Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

Applicant	CHENG-HSUAN LEE
Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Progress report
Primary Role	Team lead, test planner, and debugging coordinator
Original Source	drone-third-progress-report.pdf (45 pages)

Executive Summary

The third progress stage emphasized hardware iteration, thrust experiment interpretation, remote-control packet debugging, and final strategy development.

At this point, the project shifted from making a drone that could be assembled to making a system that could be trusted in the final evaluation.

Applicant Contribution Signals

- Linked propulsion data to mechanical and control decisions.
- Worked through remote-control packet and signal-chain uncertainty.
- Converted late-stage issues into a final strategy instead of scattered repairs.

Late-Stage Integration

The team used testing to determine which parts of the system needed redesign, reinforcement, recalibration, or procedure changes.

This stage strengthened the connection between experimental data and final mission planning.

Final Strategy

The strongest path was to simplify risk, verify core behaviors, prepare spares, and make the final attempt predictable.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Propulsion margin	Used thrust experiment results	Guided final design confidence	Experimental reasoning

Constraint	Decision	Why It Mattered	Skill Demonstrated
Remote control uncertainty	Debugged command and packet behavior	Improved control reliability	Embedded communication
Final attempt pressure	Developed practical final strategy	Reduced unnecessary risk	Execution leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.